Use Ael-Agents: Design a brain-penetrant (CNS-active), orally bioavailable allosteric covalent-reversible inhibitor and a 24-hour twice-daily (q12h BID) 2-compartment clinical PK/PD dosing regimen targeting mutant EGFR-L858R/T790M/C797S triple-mutant NSCLC with leptomeningeal brain metastases.

Used 6 tools, loaded tools, and 2 more tools

Designing a drug inhibitor and loading needed tools.

Searched available tools Ael-Agents pipeline specialist roster molecule PKPD

Planning a multi-agent drug discovery workflow with specialist hypotheses.

Ael-Agents_propose_specialist_roster

Choosing a low-polarity nitrile warhead for selective binding

Ael-Agents_cortical_rdm_and_jacobian_gate

Evaluating candidate molecules with realistic pharmacokinetic parameters.

Ael-Agents_verify_molecule_pkpd

Assessing a drug's pharmacokinetics against standard modeling assumptions.

Ael-Agents_verify_molecule_pkpd

Balancing polarity against warhead choice for CNS compliance

Ael-Agents_verify_molecule_pkpd

Evaluating a candidate molecule's drug-like properties.

Ael-Agents_dream_rsi_consolidate

Recalled memory ael-agents.md

Added to memory ael-agents.md

Assembling the final answer with diagrams and revision notes.

Brain-penetrant allosteric covalent-reversible EGFR inhibitor with a q12h

PK/PD regimen

Results up front

I ran the Ael-Agents pipeline (drug-discovery roster, Jacobian gate, Dream-RSI, RDKit and SciPy verification). This is a **computational design hypothesis**, not a validated drug.

- **Lead (AEL-EGFR-01):** `Cc1cnc(NC(=O)C(c2cc(F)ccc2O)N2Cc3cc(C=O)ccc3C2=O)s1`
- **Properties (RDKit):** MW 425.4, cLogP 3.44, HBD 2, TPSA 99.6, QED 0.61, Lipinski pass.
- **Regimen simulated:** 150 mg PO q12h. Steady-state peak:trough was 3.41, and free CSF trough was 83.8 nM, giving 94.4% occupancy.
- **Dream-RSI:** free energy fell from 0.0583 to 0.0315 (strictly improved).
- **Gate:** the Jacobian gate kept SystemsPharmacology and MedicinalChemistry as the working-memory pair. The bridge between them is the central tension: you need high CSF occupancy, but each polar warhead costs CNS penetration.

Caveats you should know:

1. **TPSA 99.6 is above the CNS-optimal window.** The tool flagged `cns_bbb_optimal: false`. I am relying on the phenol-to-amide intramolecular hydrogen bond to mask polarity. RDKit's TPSA does not credit that, so it must be tested (MDCK-MDR1 efflux).
2. **The PK output did not depend on the molecule.** Four different SMILES gave identical PK numbers. The CNS audit also uses fixed internal assumptions: MW 400, F = 1.0, fu = 0.08, Kp,uu = 0.65. The PK block is therefore a scenario simulation. F = 1.0 is optimistic, and Kp,uu is an assumption that must be measured.
3. **Potency is assumed, not measured.** I set Kd = 5 nM and IC50 = 10 nM. For a Hill coefficient of 1, IC90 = 9 × IC50 = 90 nM. Free CSF trough (83.8 nM) is about 7% below that, so coverage is marginal, not comfortable.
4. **The warhead is speculative.** An aryl aldehyde forming a reversible imine with Lys745 is plausible but unproven. Aldehydes are metabolically labile (aldehyde oxidase and ALDH oxidize them to the inactive acid).

Candidates the tool rejected:

Candidate	MW	TPSA	HBD	Verdict
Morpholinoaryl extension (no warhead)	558.6	95.0	2	Fails Lipinski (2 violations)
Boronic acid warhead	441.3	123.0	4	Fails CNS filter
Aldehyde warhead (lead)	**425.4**	**99.6**	**2**	Passes Lipinski, borderline CNS

1. Definition

An **allosteric covalent-reversible EGFR inhibitor** binds a pocket away from the ATP site and forms a reversible chemical bond with a nearby residue. It blocks the mutant kinase without needing Cys797. A **q12h two-compartment PK/PD regimen** is a twice-daily dose schedule chosen by modeling drug movement between blood and tissue, so enough drug stays on target at the trough.

2. Basics from zero

- EGFR is a receptor kinase that tells cells to grow. Mutations (L858R) keep it permanently on.
- **Resistance mutations:**
  - Early drugs failed after T790M appeared.
  - Osimertinib fixed that by bonding to **Cys797**.
  - **C797S** replaces that cysteine with serine, so the bond cannot form.
- **Leptomeningeal disease** means tumor cells float in the cerebrospinal fluid (CSF). The drug must cross the blood-brain barrier and stay above its effective concentration in CSF.

3. Intuition

Question	Answer
Problem solved	C797S removes the covalent anchor of 3rd-generation drugs
Why allosteric	The binding pocket forms when the αC-helix swings out in L858R/T790M, and it does not involve Cys797
Why reversible-covalent	It gives long residence time without the permanent haptenation risk of irreversible warheads
When used	After osimertinib failure with C797S, especially with CNS spread

4. Working principle and diagram

```
ATP site (hinge, C797S)      Allosteric pocket (αC-helix OUT)
  osimertinib blind here  -->  [phenol H-bond] [isoindolinone] [CHO---Lys745]
                                  Step 1: fast non-covalent (Ki ~ low nM)
                                  Step 2: reversible imine, k_off ~ 1e-4 /s
Effective Kd < Ki  -->  long residence  -->  sustained pEGFR suppression
```

Mutant selectivity comes from contacts with the Met790 gatekeeper and the destabilized L858R conformation. This spares wild-type EGFR, which limits rash and diarrhea.

5. Step-by-step design logic

1. **Start from the validated scaffold lineage.** Published allosteric inhibitors such as EAI045 and the JBJ compounds use an isoindolinone-thiazolylacetamide core. I recalled this from training knowledge and did not run a literature search.
2. **Remove brain-blocking features.** Skip the basic piperazine (efflux, hERG risk) and keep the molecule under about 450 Da.
3. **Block metabolism.** A 5-methyl group on the thiazole blocks CYP oxidation.
4. **Add the smallest warhead.** The aldehyde costs less TPSA than boronic acid.
5. **Verify** with RDKit, then simulate PK/PD.
6. **Falsify** with the assays below.

6. Real-world and engineering examples

- **Real-world analogy:** a Velcro hook (reversible bond) on a key (allosteric fit). It holds long but can release, unlike glue.
- **Engineering example:** osimertinib reaches the CNS and is used for brain metastases. The goal here is the same CNS reach plus C797S activity.

7. PK/PD formulas

Formulas and variables

- Average steady-state concentration: $C_{ss,avg} = F \cdot Dose / (CL \cdot \tau)$
  - F = bioavailability, CL = clearance (L/h), τ = dosing interval (h).
- Accumulation ratio: $R = 1 / (1 - e^{(-k \cdot \tau)})$
  - k = CL/Vss, with Vss = Vc + Vp (volume of distribution at steady state).
- Occupancy: $\theta = C_{free} / (C_{free} + K_d)$
- Two-compartment ODEs:
  - $dA_{gut}/dt = -ka \cdot A_{gut}$
  - $dC_c/dt = ka \cdot A_{gut}/Vc - (CL/Vc + Q/Vc) \cdot C_c + (Q/Vp) \cdot C_p$
  - $dC_p/dt = (Q/Vc) \cdot C_c \cdot (Vc/Vp) - (Q/Vp) \cdot C_p$

Use these to set a dose that keeps trough occupancy above 90%.

Worked example (inputs: 150 mg, F = 1, CL = 10 L/h, Vc = 60 L, Vp = 45 L, τ = 12 h)

1. $C_{ss,avg} = 150 / (10 \times 12) =$ **1.25 mg/L**, which is 1.25/400 = 3.125 µM, or 3125 nM.
2. Vss = 60 + 45 = 105 L, so k = 10/105 = 0.095 /h, and $t_{½} \approx 0.693/0.095 \approx$ **7.3 h**.
3. $k\tau = 1.143$, and $e^{(-1.143)} = 0.319$, so **R = 1.47**.
4. Simulated total trough = 1611 nM. Free plasma = 0.08 × 1611 = **128.9 nM**.
5. Free CSF = 0.65 × 128.9 = **83.8 nM**.
6. $\theta = 83.8 / (83.8 + 5) =$ **94.4%**, which matches the tool.

The peak:trough ratio of 3.41 is slightly above my target of 3. A modified-release formulation would flatten it.

8. Advantages and disadvantages

Advantages	Disadvantages
Works regardless of C797S	Aldehyde is a metabolic and reactivity liability
Reversible bond lowers haptenation risk	TPSA near 100 is a CNS-penetration risk
Mutant-selective, sparing wild-type EGFR	Single-protomer blockade can be bypassed by the EGFR dimer (cetuximab cannot enter the CNS)
BID fits t½ of about 7 h	Potency and Kp,uu are assumptions until measured

9. Falsifiable tests

Test	Pass criterion
Jump-dilution washout	Residence time above 2 h
MDCK-MDR1	Efflux ratio under 2
Rat Kp,uu,brain	Above 0.3
Ba/F3 L858R/T790M/C797S pEGFR	IC90 under 90 nM, with over 30-fold selectivity versus wild-type
Leptomeningeal xenograft	Unbound CSF trough above cellular IC90
Metabolic stability (hepatocytes)	Aldehyde survives; otherwise switch warhead

10. Common mistakes

- Treating simulated PK as molecule-specific.
- Judging CNS exposure by total plasma, not unbound CSF.
- Ignoring that Kp,uu must be measured.
- Forgetting MET amplification, KRAS, and HER3 bypass as resistance routes.

11. Key points for exams

C797S breaks the covalent anchor. Allosteric binding is C797-independent. Occupancy = C/(C+Kd). Accumulation R = 1/(1 − e^(−kτ)). The CNS rule of thumb is MW below 450, TPSA 40–90, HBD ⩽ 2.

This is preclinical design work only. Any human dose would come from toxicology and phase 1 data, not this simulation.

Quick revision notes

- Allosteric pocket = αC-out, independent of Cys797.
- Reversible-covalent gives long residence without permanent adducts.
- Lead MW 425, TPSA 99.6, HBD 2. CNS fit is borderline.
- Simulated 150 mg BID gives 94.4% free-CSF trough occupancy, but IC90 coverage is marginal.
- PK numbers come from fixed assumptions, so measure Kp,uu, F, and fu.

5 viva questions

1. Why does C797S defeat osimertinib but not an allosteric inhibitor?
2. What does “reversible covalent” mean, and why prefer it?
3. Why does unbound CSF concentration drive leptomeningeal efficacy?
4. Define occupancy and write its formula.
5. Why was the basic piperazine removed?

5 interview questions

1. How would you decide between boronic acid and aldehyde warheads?
2. TPSA is 99.6. How do you de-risk BBB penetration experimentally?
3. How do you convert a PK simulation into a first-in-human dose?
4. Which resistance mechanisms would you expect after this drug?
5. Why might dimerization limit single-agent allosteric inhibitors?

5 MCQs

1. C797S mainly blocks which drug feature? (a) Hinge binding (b) Covalent bond to Cys797 (c) Solubility (d) Absorption. **Answer: b**
2. Occupancy at C_free = 83.8 nM and Kd = 5 nM is closest to: (a) 50% (b) 75% (c) 94% (d) 100%. **Answer: c**
3. Accumulation ratio with k = 0.095/h and τ = 12 h is about: (a) 1.0 (b) 1.5 (c) 3.0 (d) 5.0. **Answer: b**
4. Which property most limits CNS entry here? (a) MW 425 (b) QED 0.61 (c) TPSA 99.6 (d) cLogP 3.44. **Answer: c**
5. Why did the PK output not change between SMILES? (a) Molecules are identical (b) The tool’s PK uses fixed assumptions plus numeric inputs (c) Dose differed (d) RDKit failed. **Answer: b**

If you want, I can rerun the PK simulation at other doses, or test other warheads (such as a benzoxaborole) against the CNS filter.